Natural Language Inference: A Comparative Study of Classical and Transformer-Based Approaches

Part 2 Report

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Abstract

Natural Language Inference (NLI) asks whether a hypothesis follows from a premise, contradicts it, or is left undetermined by it. This report compares classical machine-learning baselines with a fine-tuned Transformer on the Stanford Natural Language Inference (SNLI) corpus. After removing records with invalid labels, the data comprise 549,367 training, 9,842 validation and 9,824 test pairs. To keep computation manageable and to give every model identical training data, all experiments use one stratified subset of 50,000 training examples. Logistic Regression, LinearSVC and LightGBM are trained on TF-IDF and engineered pairwise features, and DistilBERT is fine-tuned for three-way classification for a single epoch. Logistic Regression is the strongest classical model on validation data (72.30% accuracy) and reaches 71.86% on the test set. DistilBERT obtains 81.91% validation accuracy and 81.90% test accuracy, with a test macro-F1 of 81.88%. Confusion matrices show that confusions involving the neutral class remain common, although DistilBERT cuts the number of test errors from 2,764 to 1,778. Experiments with large language models and evaluation on MultiNLI are planned for Part 3.

Scope of this report. This document covers the completed work on SNLI exploratory analysis, classical baselines and DistilBERT fine-tuning. LLM-based approaches and generalisation to MultiNLI are planned for Part 3 and are not presented here as completed experiments.

1 Introduction

Natural Language Inference, also known as recognising textual entailment, is the task of predicting the relationship between two pieces of text. The first sentence is called the premise and the second the hypothesis. A hypothesis is labelled *entailment* if it follows from the premise, *contradiction* if it conflicts with the premise, and *neutral* if the premise gives too little information to decide. For instance, the premise "A man camps outside with a tent" supports the hypothesis "The man is outside", but it says nothing about why the man is camping or about other details.

NLI is a useful probe of language understanding because it cannot be solved by matching words alone. A system has to account for meaning, negation, missing information and the relation between two sentences. These abilities matter for fact-checking, question answering and consistency checking of generated text. Strong benchmark scores, however, do not by themselves show robust reasoning. Earlier work found that a classifier can sometimes reach good accuracy by exploiting statistical cues in the hypothesis while ignoring the premise [1]. For this reason, the models studied here are treated as baselines, and the planned cross-genre evaluation is expected to say more about how well they generalise.

The aim of this project is to compare three classical approaches (Logistic Regression, LinearSVC and LightGBM) with a fine-tuned Transformer (DistilBERT) on the same SNLI task. Part 3 will add LLM-based methods and test generalisation on MultiNLI, whose wider range of genres makes it a more demanding setting.

2 Literature Survey

2.1 NLI datasets and the task

Bowman et al. [2] introduced SNLI, a large human-labelled corpus of sentence pairs for three-way inference. Its premises are drawn mainly from image captions, so most examples describe short, concrete scenes. SNLI is well suited to comparing models under a standard split, but its domain is narrow compared with the text encountered in practice. Williams et al. [3] later released MultiNLI, which contains examples from ten genres of written and spoken English, including government documents, fiction and conversation. Their work motivates measuring not only in-domain accuracy but also how performance changes when the genre changes.

2.2 Classical machine-learning approaches

TF-IDF represents a document through term weights that reflect how informative each term is across a corpus. Combined with unigrams and bigrams, it gives an efficient baseline that captures lexical overlap and some local word order. Logistic Regression models class probabilities from a weighted feature vector, while a linear Support Vector Machine learns a decision boundary by maximising a margin [4]. Both train quickly and are relatively easy to inspect, but neither builds contextual token representations the way a Transformer does. LightGBM is a gradient-boosted decision-tree framework designed for efficient training [5]. In this project it operates on compact bag-of-words and engineered pairwise features rather than on the full sparse TF-IDF representation used by the linear models.

2.3 Transformer-based inference

BERT pretrains a bidirectional Transformer encoder on large text collections and then fine-tunes it for downstream tasks. Devlin et al. [6] showed strong results across language-understanding benchmarks, NLI among them. DistilBERT was produced through knowledge distillation to reduce model size and computation while keeping much of BERT's accuracy [7], which makes it a practical choice when compute is limited. Here, the pretrained `distilbert-base-uncased` checkpoint is extended with a three-class classification head and fine-tuned on SNLI.

2.4 Position of the current study

This phase establishes a controlled baseline comparison with one shared 50,000-example training subset and fixed random seeds. The gap between the sparse-feature models and DistilBERT offers initial evidence for the value of contextual representations on SNLI. It is not yet a complete comparison, since LLMs and domain-shifted data are still to be examined. Those experiments, together with a closer look at individual misclassified examples, are left to Part 3.

3 Dataset and Preprocessing

3.1 Dataset source and structure

The project uses SNLI as distributed on Hugging Face under the identifier `stanfordnlp/snli` [8]. Each record holds a premise, a hypothesis and a label, with the mapping 0 = entailment, 1 = neutral and 2 = contradiction. The loader provides 550,152 training, 10,000 validation and 10,000 test rows. Some records carry an unresolved label (-1), and these were removed before training and evaluation. Table 1 gives the counts.

Table 1: Cleaning counts from the Kaggle execution log.

Split	Rows before cleaning	Invalid labels removed	Rows after cleaning
Train	550,152	785	549,367
Validation	10,000	158	9,842
Test	10,000	176	9,824
Total	570,152	1,119	569,033

The exploratory analysis script checks for missing or empty sentences, duplicate premise–hypothesis pairs, unique premises and hypotheses, label counts, sentence lengths, vocabulary coverage and token lengths. After cleaning, the log reports no null or empty premise or hypothesis fields in any split. The training split contains 626 duplicate pairs, the validation split has 2, and the test split has none.

3.2 Class distribution and initial observations

As Table 2 and Figure 1 show, the three classes are close to balanced in every split.

Table 2: Class counts and percentages after cleaning.

Split	Entailment	Neutral	Contradiction
Train (549,367)	183,416 (33.39%)	182,764 (33.27%)	183,187 (33.35%)
Validation (9,842)	3,329 (33.82%)	3,235 (32.87%)	3,278 (33.31%)
Test (9,824)	3,368 (34.28%)	3,219 (32.77%)	3,237 (32.95%)

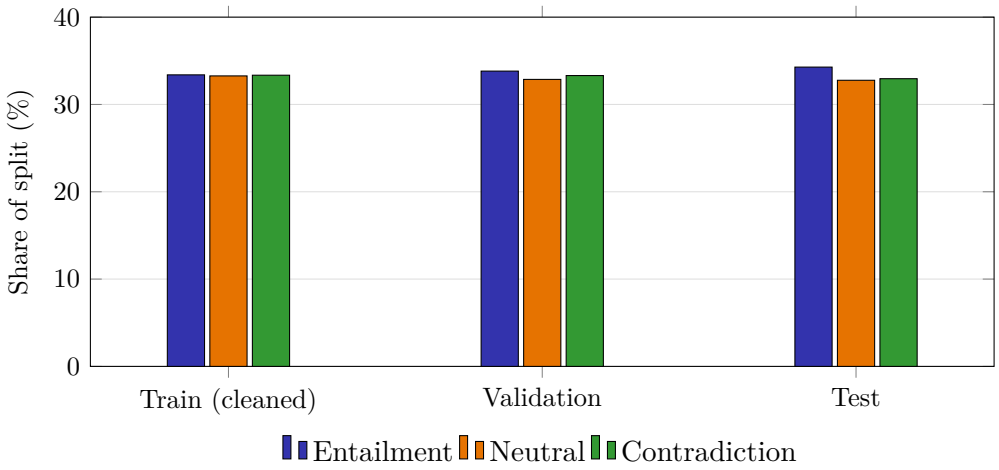


Figure 1: Label percentages are broadly balanced across the training, validation and test splits.

Table 3 summarises the main exploratory findings. Premises average 12.85 words and hypotheses 7.42 words, so most pairs are short. Lexical overlap with the premise falls from entailment to contradiction, which makes it a useful but incomplete cue. Negation in the hypothesis is more frequent for contradiction, but at 4.9% it is far from a reliable rule on its own.

Table 3: Selected exploratory data analysis findings from the captured log.

Statistic	Observed result	Interpretation
Mean train premise length	12.85 whitespace-separated words	Premises are generally short scene descriptions.
Mean train hypothesis length	7.42 words	Hypotheses are shorter than premises on average.
95th percentile length (train)	Premise: 23 words; hypothesis: 13 words	Most pairs are short, though longer examples exist.
Training vocabulary	33,710 word types; 11,160,583 tokens	Lexical coverage is large, but repeated words dominate token counts.
OOV token rate	Validation: 0.196%; test: 0.195%	Very few evaluation tokens are unseen by the simple EDA tokenizer.
Hypothesis-word coverage	Entailment: 57.0%; neutral: 42.1%; contradiction: 38.2%	Overlap varies by class and may be a useful but incomplete cue.
Hypothesis negation	Entailment: 0.8%; neutral: 0.9%; contradiction: 4.9%	More common in contradiction, but not a reliable rule by itself.

A small number of premise texts occur in more than one of the original splits: 16 are shared between train and validation, 11 between train and test, and 1 between validation and test. Exact pair overlap was one pair between train and validation and none for the other comparisons. The sampled 50,000-example training subset had no exact pair overlap with the validation or test sets. This is reassuring with respect to exact-pair leakage, but the small amount of shared premise text in the full splits remains a dataset caveat.

3.3 Preprocessing

Cleaning keeps only records whose label is in $\{0, 1, 2\}$, converts missing sentence values to empty strings where present, and saves only the premise, hypothesis and label columns. No stemming

or global stop-word removal is applied. This is deliberate, because function words such as "not" and "no" can change the inference relation. The classical pipeline lowercases the text and tokenises with the pattern `[a-z0-9]+(?:'[a-z]+)?`, which keeps alphanumeric tokens and common contractions.

For the linear models, separate TF-IDF vectors are built for the premise and hypothesis. These are supplemented with absolute vector differences, a cosine-like overlap feature, TF-IDF features for hypothesis words that are absent from the premise, standardised numeric pair features and hashed cross-pair features. The main vectoriser uses up to 50,000 features, unigrams and bigrams, `min_df=2`, sublinear term frequency and L2 normalisation. Together these features capture lexical similarity, novel hypothesis content, sentence lengths and negation flags. LightGBM uses a smaller matrix made up of engineered numeric features, cosine similarity, binary hypothesis bag-of-words features and binary features for hypothesis words missing from the premise.

For DistilBERT, the Hugging Face tokenizer receives the premise and hypothesis as a text pair, adds the model-specific special tokens, truncates sequences to 128 tokens and pads batches dynamically. This is contextual tokenisation and involves no manual stop-word removal or stemming. Table 4 shows what the classical analyzer produces for one example.

Table 4: Lowercased unigram and bigram features produced by the TF-IDF analyzer for one entailment example. The tokens shown are not the final weighted numeric vector.

Raw example	Classical token-feature view
Premise: "Three people hiking up a mountain with a river and other mountains in the background." **Hypothesis:** "Hikers in the mountains."	**Premise:** three, people, hiking, up, mountain, river, background, three people, people hiking, hiking up, ... **Hypothesis:** hikers, in, the, mountains, hikers in, in the, the mountains

4 Evaluation Metrics

Models are evaluated as three-class classifiers using accuracy, macro-F1, weighted-F1 and class-wise precision, recall and F1. Accuracy is the fraction of examples given the correct label. Macro-F1 computes F1 separately for each class and averages the three values with equal weight, so performance on every relation stays visible. Weighted-F1 averages the class F1 values in proportion to class support. Because SNLI is nearly balanced, the two are similar, but reporting both makes the evaluation more complete. Precision is the proportion of predictions for a class that are correct, and recall is the proportion of true examples of that class that are recovered. The confusion matrices in Section 6 show which class pairs are confused most often, while a manual review of individual errors would be needed to explain why.

5 Models and Experimental Setup

5.1 Shared split and reproducibility

The classical training script draws a single stratified random sample of 50,000 rows from the cleaned SNLI training split using seed 42. The same file is reused for DistilBERT, so every reported model is trained on identical data. The subset contains 16,693 entailment, 16,634 neutral and 16,673 contradiction examples. Validation uses all 9,842 cleaned examples and

testing uses all 9,824. Preprocessing components are fitted on the training data, the best classical model is selected by validation accuracy, and only that model is then evaluated on the test set. The test split is therefore not used for classical model selection.

5.2 Classical models

Logistic Regression and LinearSVC share a high-dimensional sparse representation, whereas LightGBM trains boosted trees on compact lexical and numeric features. Each model runs with a fixed configuration, and no hyperparameter grid search was performed (Table 5). Model fitting took about 178 seconds for Logistic Regression, 65 seconds for LinearSVC and 17 seconds for LightGBM in the logged Kaggle run. These are fitting times and not necessarily the end-to-end times of the full pipeline.

Table 5: Classical models and main settings from the training script.

Model	Feature representation	Main hyperparameters
Logistic Regression	Sparse TF-IDF pair features plus engineered and hashed features	$C = 1.0$; lbfgs; `max_iter`=1000; tol=10^{-4}; seed 42
LinearSVC	Same linear feature matrix as Logistic Regression	$C = 0.1$; L2 penalty; squared-hinge loss; `max_iter`=2000; tol=10^{-4}; seed 42
LightGBM	Numeric pair features, cosine similarity, binary hypothesis and novel-word features	Multiclass; up to 1,500 estimators; learning rate 0.08; 63 leaves; `min_child_samples`=40; subsample 0.8; `colsample_bytree`=0.6; L2 $\lambda = 1$; early stopping after 50 rounds

5.3 DistilBERT

The Transformer experiment uses `distilbert-base-uncased` with a sequence-classification head for three labels. The encoder is fine-tuned on the premise–hypothesis pair, so its token representations are updated for NLI and not kept frozen. The selected configuration uses one epoch, a learning rate of 5×10^{-5}, a maximum sequence length of 128, a per-device batch size of 8 with 8 gradient-accumulation steps (effective batch size 64), weight decay of 0.01, a linear learning-rate schedule and 10% warm-up. The seed and data seed are both 42. Training took approximately 4.4 minutes.

5.4 Software and compute

The scripts use Python, pandas, NumPy, scikit-learn, SciPy, LightGBM, Hugging Face Datasets and Transformers, PyTorch and joblib. Training was run through Kaggle notebooks. The available log does not identify the accelerator model, so no GPU type is claimed. The DistilBERT script selects bfloat16 automatically when a supported CUDA device is present, fp16 on other CUDA devices, and fp32 without CUDA.

6 Results and Discussion

6.1 Validation and test results

Table 6 reports results for all completed experiments. The classical script evaluates only the validation-selected Logistic Regression model on the test set, so LinearSVC and LightGBM have no reported test scores.

Table 6: Validation and test results for the completed experiments.

Model	Val. acc.	Val. macro-F1	Training time	Test acc.	Test macro-F1
Logistic Regression	0.7230	0.7226	178 s (fit)	0.7186	0.7181
LinearSVC	0.7217	0.7207	65 s (fit)	n/a	n/a
LightGBM	0.7043	0.7035	17 s (fit)	n/a	n/a
DistilBERT	**0.8191**	**0.8189**	4.4 min	**0.8190**	**0.8188**

Among the classical models, Logistic Regression has the highest validation accuracy at 72.30%, narrowly ahead of LinearSVC at 72.17%. LightGBM reaches 70.43% but is the fastest to fit, at 17 seconds against 65 for LinearSVC and 178 for Logistic Regression. Logistic Regression was selected on validation accuracy and then evaluated once on the held-out test set, where it obtained 71.86% accuracy, 0.7181 macro-F1 and 0.7188 weighted-F1. This procedure keeps the test set out of the choice among classical models.

On the same test set, DistilBERT improves accuracy by 10.04 percentage points over the selected Logistic Regression baseline (81.90% versus 71.86%). Test macro-F1 rises by 10.07 points (0.8188 versus 0.7181), and weighted-F1 is 0.8193 versus 0.7188. DistilBERT's validation and test accuracies are almost identical (81.91% and 81.90%), which is encouraging. Even so, a single run on one training subset cannot establish stability across random seeds or larger training samples. Training took about 4.4 minutes and test inference 18.4 seconds in the logged Kaggle run.

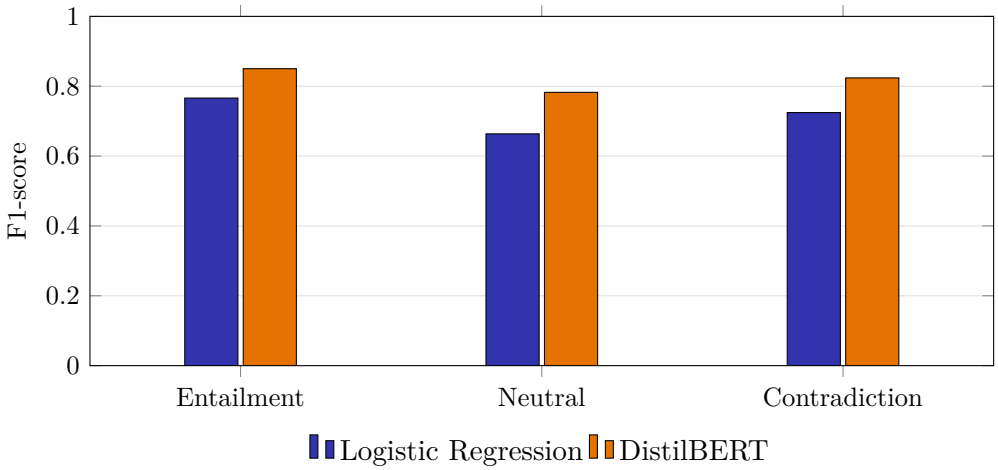


Figure 2: Test-set F1 by NLI class for the two models with test predictions in the captured log.

6.2 Class-wise performance

Neutral is the weakest class for both models by F1 (Table 7 and Figure 2). Logistic Regression scores 0.6636 on neutral, compared with 0.7661 on entailment and 0.7246 on contradiction. DistilBERT improves all three classes, reaching 0.8502 on entailment, 0.7825 on neutral and 0.8238 on contradiction. The largest absolute gain is on neutral (+0.1189), followed by contradiction (+0.0992) and entailment (+0.0841). Neutral examples typically add information that the premise neither supports nor contradicts, so the decision depends on what the premise does and does not establish.

Table 7: Class-wise test metrics for Logistic Regression (LR) and DistilBERT (DB).

Test class	LR prec.	LR rec.	LR F1	DB prec.	DB rec.	DB F1
Entailment	0.7626	0.7696	0.7661	0.8438	0.8566	0.8502
Neutral	0.6557	0.6716	0.6636	0.7625	0.8037	0.7825
Contradiction	0.7372	0.7124	0.7246	0.8546	0.7952	0.8238

6.3 Confusion-matrix error analysis

In the confusion matrices below, rows give the gold label and columns the predicted label, so correct classifications lie on the diagonal. On the 9,824-example test set, Logistic Regression classifies 7,060 examples correctly and makes 2,764 errors. DistilBERT classifies 8,046 correctly and makes 1,778 errors. That is 986 additional correct predictions, a reduction in test errors of about 35.7% relative to Logistic Regression in this run.

Table 8: Logistic Regression confusion matrix on the SNLI test set (rows: true label; columns: predicted label).

True label	Pred. Entailment	Pred. Neutral	Pred. Contradiction
Entailment	2,592	513	263
Neutral	498	2,162	559
Contradiction	309	622	2,306

Table 9: DistilBERT confusion matrix on the SNLI test set (rows: true label; columns: predicted label).

True label	Pred. Entailment	Pred. Neutral	Pred. Contradiction
Entailment	2,885	346	137
Neutral	331	2,587	301
Contradiction	203	460	2,574

For Logistic Regression, the largest single error flow is true contradiction predicted as neutral (622 cases), followed by true neutral predicted as contradiction (559) and true entailment predicted as neutral (513). The boundary between neutral and contradiction is therefore a major source of error for this model. DistilBERT reduces these counts to 460, 301 and 346 respectively. Its largest remaining off-diagonal cell is again contradiction predicted as neutral (460), and errors involving neutral stay prominent. This agrees with the class-wise F1 results: neutral is still DistilBERT's lowest-scoring class, even though its gain over Logistic Regression is the largest.

Table 10 lists dataset examples that illustrate challenges such as unstated intent, negation and paraphrase. They are included to explain the task and are not verified instances of these models' errors. A future instance-level review of the prediction files would help connect individual sentence phenomena to the aggregate patterns reported here.

Table 10: Illustrative SNLI examples of inference challenges. These are dataset examples, not confirmed model mispredictions.

Challenge	Example	Why it matters
Unstated intention or cause	**P:** "Brown dog jumping off of a deck." **H:** "The dog is jumping off a deck to save his master." **Label:** neutral	The hypothesis adds a motive that the premise does not establish.
Negation	**P:** "Three children are sitting around a table reading books." **H:** "None of the kids are reading." **Label:** contradiction	A system must represent the scope and meaning of negation, not only word overlap.
Paraphrase and broader meaning	**P:** "Three people hiking up a mountain with a river and other mountains in the background." **H:** "Hikers in the mountains." **Label:** entailment	The hypothesis changes the wording and drops detail while preserving the main meaning.

7 Limitations and Next Steps

- **Training subset.** The models use 50,000 of the 549,367 cleaned training examples. This keeps experiments practical, but results may change with more data.
- **Limited tuning.** The classical configurations are fixed and DistilBERT was trained for one epoch. Multiple seeds and controlled hyperparameter tuning have not yet been tried.
- **Incomplete test comparison.** Test metrics are reported only for the validation-selected classical model and for DistilBERT. LinearSVC and LightGBM have validation metrics only.
- **Qualitative error review.** Confusion matrices and class-wise metrics are available, but a manual instance-level review of confirmed mispredictions has not been done.
- **Domain limitations.** SNLI is largely based on image-caption descriptions. The planned MultiNLI evaluation will test whether the models transfer to other genres without additional fine-tuning.
- **Part 3 experiments.** Implement LLM-based baselines (prompt-based and/or parameter-efficient fine-tuning, depending on available compute), evaluate them on the same SNLI test set where feasible, and test SNLI-trained models on MultiNLI. Compare accuracy, macro-F1, per-class performance and compute cost.

8 Conclusion

This phase establishes a reproducible NLI pipeline that runs from SNLI cleaning and exploratory analysis through three classical models and a fine-tuned DistilBERT. The cleaned dataset is close to class-balanced and its sentence pairs are generally short. Logistic Regression is the strongest classical model by validation accuracy and reaches 71.86% test accuracy. DistilBERT achieves 81.90% test accuracy and 0.8188 macro-F1, an improvement of 10.04 percentage points on the shared test set. The confusion matrices show that confusions involving the neutral class are still common, particularly contradiction predicted as neutral. These results provide useful baselines for what follows. Prompted or fine-tuned LLMs, a deeper instance-level error review and generalisation tests on MultiNLI remain planned for Part 3.

A Reproducibility Artifacts

The scripts save cleaned splits, metrics, model checkpoints and per-example test predictions. The captured Kaggle log confirms that the exploratory analysis, the classical baselines and DistilBERT fine-tuning all completed successfully. The confusion matrices in Tables 8 and 9 are based on the class-level prediction counts supplied for this report. The per-example prediction files remain useful for a later instance-level review.

Table 11: Output artifacts expected from the supplied scripts.

Output artifact	Purpose
`results/cleaned/train.csv`, `validation.csv`, `test.csv`	Cleaned SNLI splits after removing invalid labels.
`results/cleaned/train_50k_seed42.csv`	Shared stratified training subset used by all four completed models.
`results/classical_results.json`	Classical validation results, selected-model test metrics and timing.
`results/distilbert_results.json`	DistilBERT configuration, validation and test metrics, and training and inference time.
`results/classical_test_predictions.csv`, `results/distilbert_test_predictions.csv`	Per-example labels and predictions for confusion matrices and qualitative error review.
`results/eda_summary.json` and EDA plots	Data integrity, length, class and overlap statistics with visual summaries.

References

[1] Poliak, A., Naradowsky, J., Haldar, A., Rudinger, R., and Van Durme, B. (2018). Hypothesis only baselines in natural language inference. In *Proceedings of the Seventh Joint Conference on Lexical and Computational Semantics*, pages 180–191. https://doi.org/10.18653/v1/S18-2023

[2] Bowman, S. R., Angeli, G., Potts, C., and Manning, C. D. (2015). A large annotated corpus for learning natural language inference. In *Proceedings of the 2015 Conference on Empirical Methods in Natural Language Processing*, pages 632–642. https://doi.org/10.18653/v1/D15-1075

[3] Williams, A., Nangia, N., and Bowman, S. R. (2018). A broad-coverage challenge corpus for sentence understanding through inference. In *Proceedings of NAACL-HLT 2018*, pages 1112–1122. https://doi.org/10.18653/v1/N18-1101

[4] Cortes, C. and Vapnik, V. (1995). Support-vector networks. *Machine Learning*, 20:273–297. https://doi.org/10.1007/BF00994018

[5] Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., Ye, Q., and Liu, T.-Y. (2017). LightGBM: A highly efficient gradient boosting decision tree. In *Advances in Neural Information Processing Systems*, volume 30.

[6] Devlin, J., Chang, M.-W., Lee, K., and Toutanova, K. (2019). BERT: Pre-training of deep bidirectional Transformers for language understanding. In *Proceedings of NAACL-HLT 2019*, pages 4171–4186. https://doi.org/10.18653/v1/N19-1423

[7] Sanh, V., Debut, L., Chaumond, J., and Wolf, T. (2019). DistilBERT, a distilled version of BERT: Smaller, faster, cheaper and lighter. *arXiv preprint arXiv:1910.01108*. `https://arxiv.org/abs/1910.01108`

[8] Stanford Natural Language Processing Group (n.d.). The Stanford Natural Language Inference (SNLI) Corpus. `https://nlp.stanford.edu/projects/snli/`